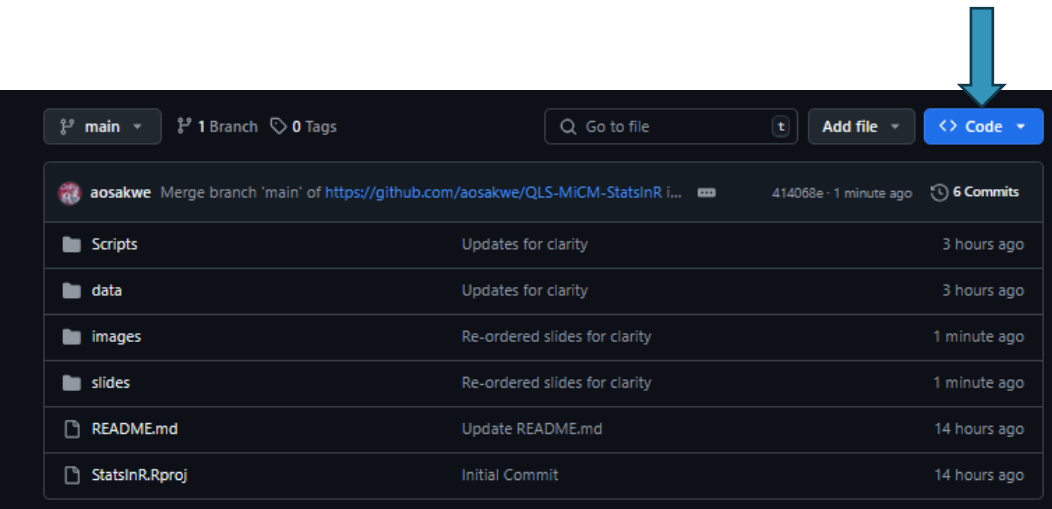


Workshop materials

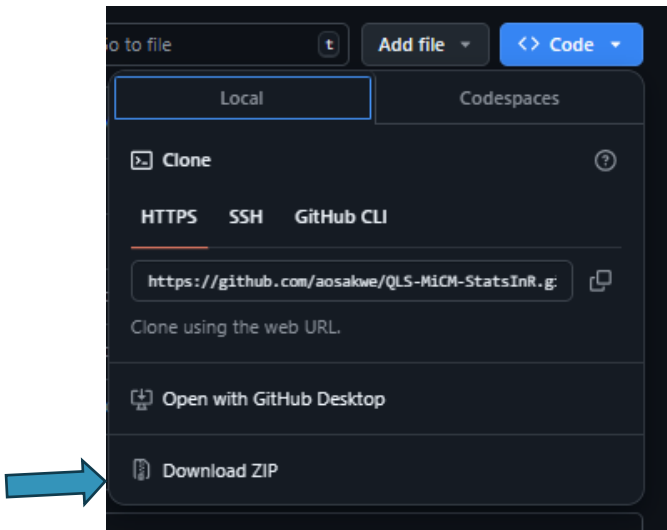
<https://github.com/QLS-MiCM/StatsInR>

use 'git clone' command OR



The screenshot shows the GitHub repository page for 'StatsInR'. At the top, there are tabs for 'main', '1 Branch', and '0 Tags'. Below this is a search bar and buttons for 'Add file' and '<> Code'. A blue arrow points to the 'Code' button. Below the buttons, there is a table of files and folders:

File/Folder	Description	Time
Scripts	Updates for clarity	3 hours ago
data	Updates for clarity	3 hours ago
images	Re-ordered slides for clarity	1 minute ago
slides	Re-ordered slides for clarity	1 minute ago
README.md	Update README.md	14 hours ago
StatsInR.Rproj	Initial Commit	14 hours ago



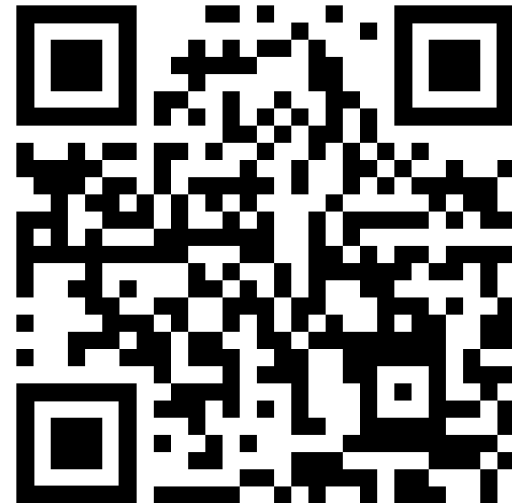
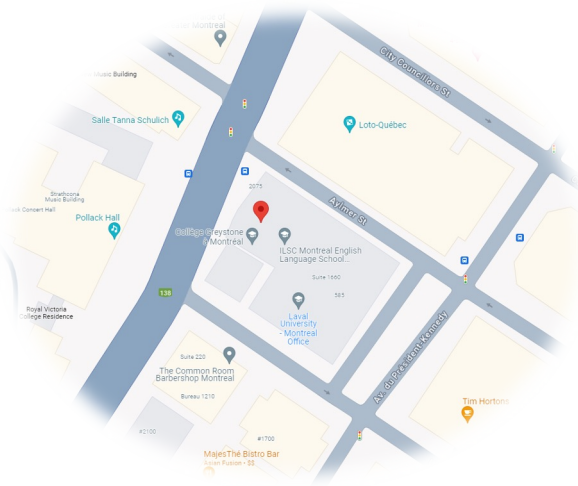
The screenshot shows the 'Code' dropdown menu. It has tabs for 'Local' and 'Codespaces'. Under 'Local', there is a 'Clone' button. Below 'Clone', there are three options: 'HTTPS', 'SSH', and 'GitHub CLI'. The 'HTTPS' option is selected, and the URL 'https://github.com/aosakwe/QLS-MiCM-StatsInR.g...' is shown. Below the URL, there is a button 'Clone using the web URL'. Below that, there is a button 'Open with GitHub Desktop'. At the bottom, there is a button 'Download ZIP'. A blue arrow points to the 'Download ZIP' button.

Statistical Analysis in R

Workshop Lead: Xiaoqi Xie
June 19, 2026

**Slides are adapted from slides
created by Adrien Osakwe for this
workshop in May.**

QLS-MiCM mission statement: deliver quality workshops designed to help biomedical researchers develop the skills they need to succeed.



Location: 550 Sherbrooke
Street, Montreal, Quebec

Scan the QR code to sign up
for our **mailing list**

Contact: workshop-micm@mcgill.ca

Workshop Series

Winter Series

Workshop	Date	Location	Registration
How to think in Code	Feb. 2 1PM-3PM	Sherbrooke 550 Room 189	Closed
Intro to Git & GitHub	Feb. 3 1PM-5PM	Sherbrooke 680 Room 1047	Closed
Intro to Unix	Feb. 5 1PM-3PM	Sherbrooke 680 Room 1047	Closed
Intro to R (Part 1)	Feb. 9 1PM-5PM	Sherbrooke 550 Room 189	Closed
Intro to Python (Part 1)	Feb. 11 1PM-5PM	Sherbrooke 680 Room 1047	Closed
Intro to MATLAB	Feb. 23 1PM-5PM	Sherbrooke 550 Room 189	Closed
Statistics in R (Part 2)	Mar. 10 1PM-5PM	Sherbrooke 680 Room 1047	Open
Data Processing in Python (Part 2)	Mar. 11 1PM-5PM	Sherbrooke 680 Room 1261	Open
Polygenic Risk Scores (PRS)	Mar. 13 1PM-5PM	Sherbrooke 680 Room 1261	Open
Data Processing for Genetics	Mar. 17 1PM-5PM	Sherbrooke 680 Room 1047	Open
Intro to Machine Learning	Mar. 23 1PM-5PM	Sherbrooke 550 Room 189	Open
More workshops to be announced soon!			

<https://www.mcgill.ca/micm/training/workshops-series>

Outline

1. Data Wrangling
2. Linear Regression Models
3. Logistic Regression Models
4. Statistical Testing
5. Study Design

Acknowledgements

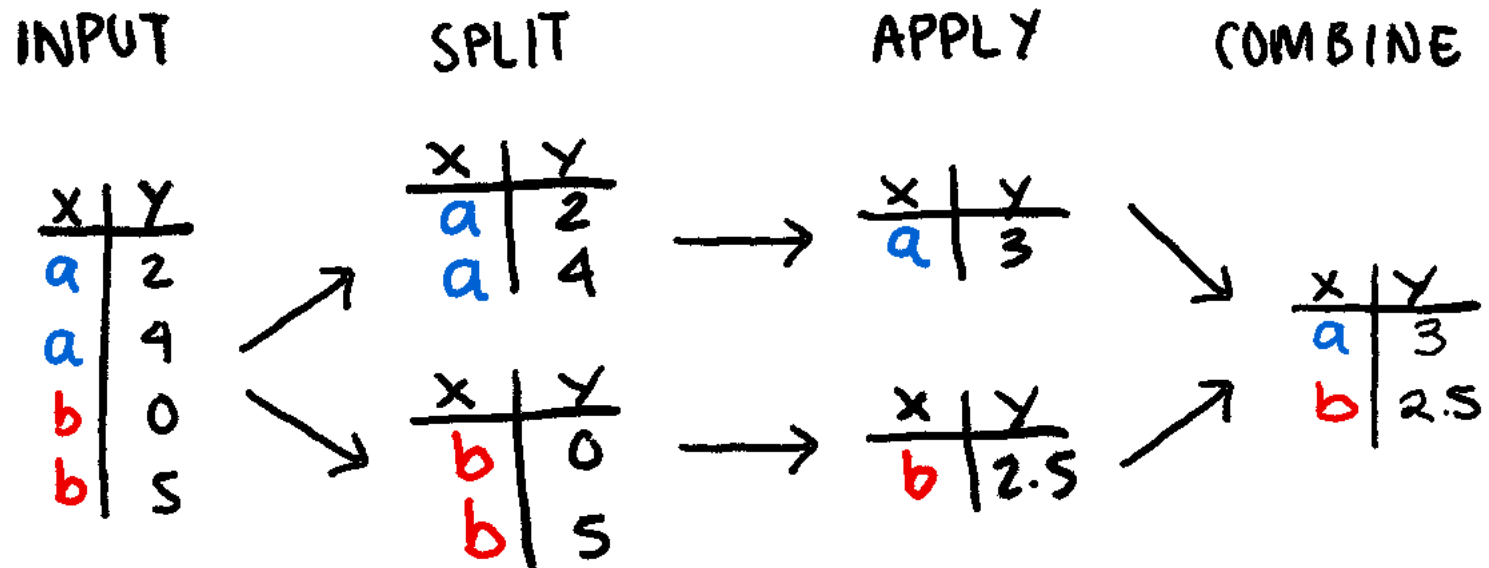
Gerardo Martinez - McGill
Alex Diaz-Papkovich - Brown
Larisa Morales Soto - HMS
Lisa Sullivan BUSPH

Data Wrangling

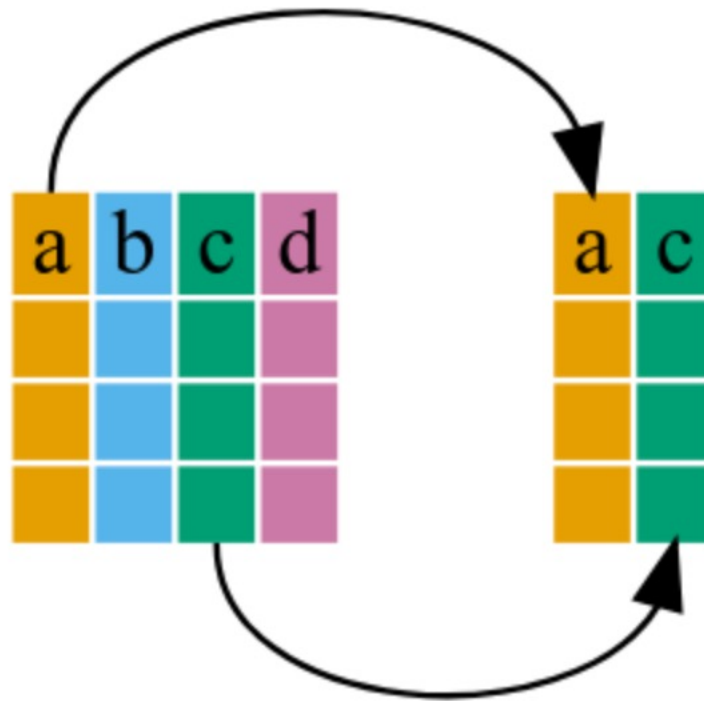
Learning objectives:

- Become familiar with the dplyr syntax
- Create pipes with the operator `%>%`
- Perform operations on data frames using dplyr and tidyr functions
- Examples for how to deal with missing data

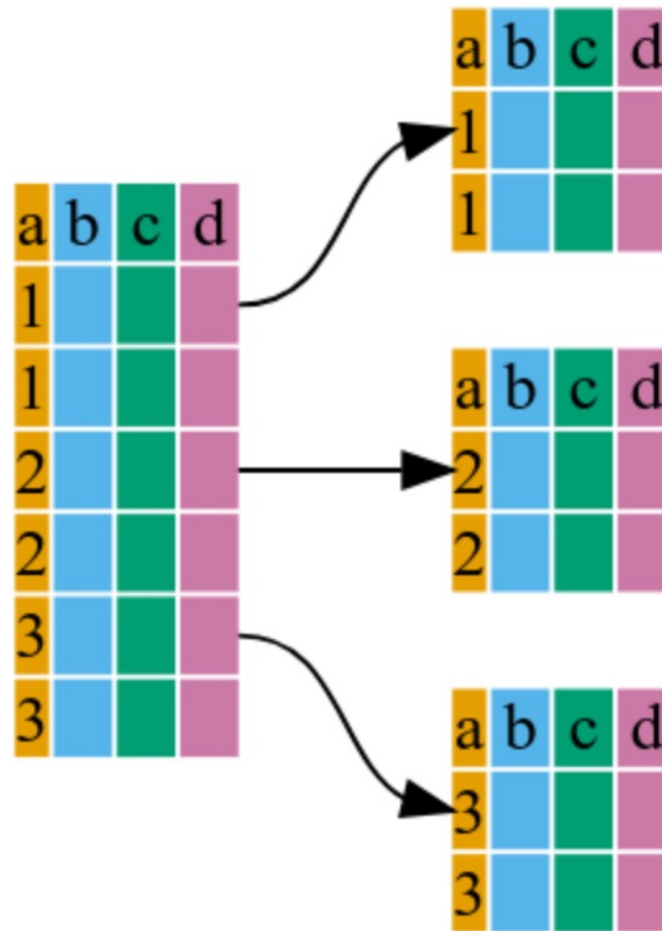
Split-Apply-Combine problem



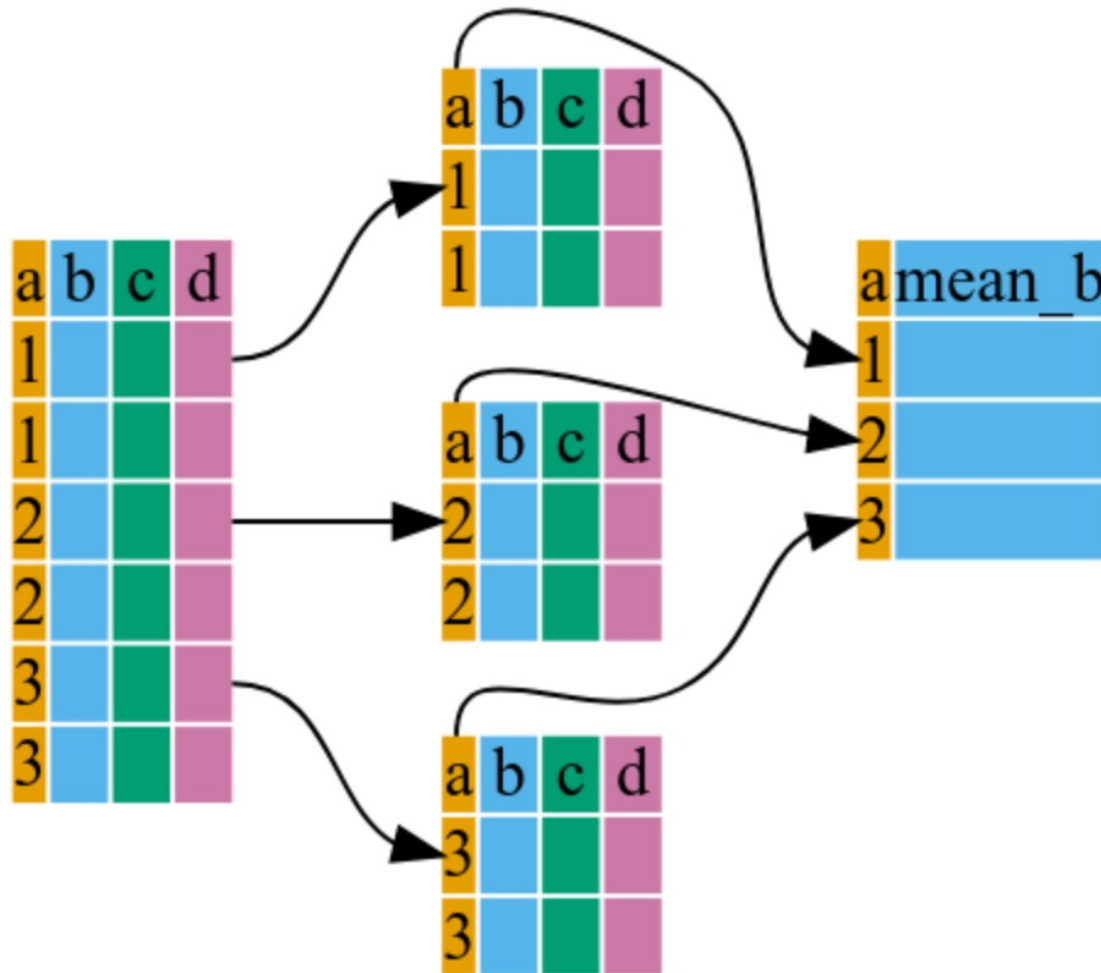
Select



Group by



Summarize



Managing Missing Data

- Most of the time, samples having missing entries
 - Missed medical appointment
 - Error with measuring instruments etc.
 - Represented as **NA** in R

Simple Solution

- Drop samples with missing observations

More rigorous solutions

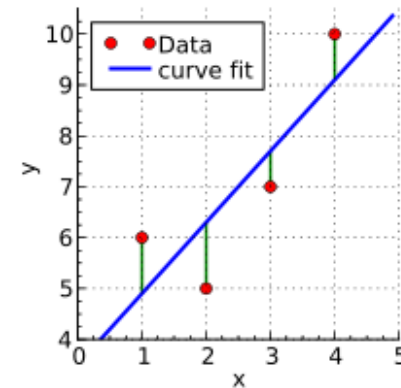
- Fill in with average/median of dataset or condition
- Train a model to 'impute' missing entries based on other features

Linear Regression

Learning objectives:

- Understand the fundamental regression algorithms
- What is a coefficient/effect size

The Classic Normal Linear Model



Credit: Wikipedia

- The fundamental model for statistics & machine learning
- Follow a **Normal distribution**

Imagine a dataset with a dependent variable y and a set of descriptive features x

We want to learn what features in x are good predictors of y (what is their **effect size/coefficient** β)

$$y = \beta x + \epsilon \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$E(y|x) = \beta x \quad y|x = \mathcal{N}(\beta x, \sigma^2)$$

Model Assumptions

1. Outcome is **continuous** and a **linear combination of predictors**

2. Outcome is such that $y_i|x_i = \mathcal{N}(\beta x_i, \sigma^2)$

3. Predictors must not be perfectly correlated (linear combination) -----
avoid multicollinearity

Why?

$$\hat{\beta} = (x^T x)^{-1} x^T y$$

4. For every observation i , the error is:

If columns of x are linearly dependent on each other, $x^T x$ is non-invertible!

1. **Normally distributed**

2. **Mean zero**

3. **Homoskedastic** (same variance as other observations)

4. **Independent** (not correlated)

Model Fitting

Probabilistic Approach

- **Maximum Likelihood Estimation**

$$\ell(\beta, \sigma^2 \mid y, X) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (y - X\beta)^\top (y - X\beta)$$

$$\hat{\beta} = \arg_{\beta} \ell$$

Machine Learning

- **Ordinary Least Square**

$$y = \beta x + \epsilon$$

$$\hat{y} = \hat{\beta} x$$

$$MSE = \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{N}$$

Extensions

More than Normal distribution! However, if we lose the normality for error term (residual), we will fail to do hypothesis tests.

Interaction effects

- Outcomes depend on interactions of (e.g.) age*sex

Popular Models for Different Types of Data

- **Multivariate regression (multiple, correlated outcomes)**
- **Logistic regression (Binary outcomes like healthy vs. disease)**
- Multinomial/Poisson regression (Count data)
- ARIMA (Time series—data correlated over time)

Logistic Regression

Learning objectives:

- What is a logistic function
- How LR extends the linear model
- What to do when we have more than 2 categories

Linear Regression could be used to predict counts (technically)
BUT it has many flaws

- Formulation isn't bounded (0,1)
- Impractical for predicting categories

If we have binary outcomes (0 or 1), logistic regression is a better option!

Logistic Regression (Generalized Linear Model)

- Sigmoid/Logistic function provides us with a bounded output
- Range is also easy to interpret as a probability

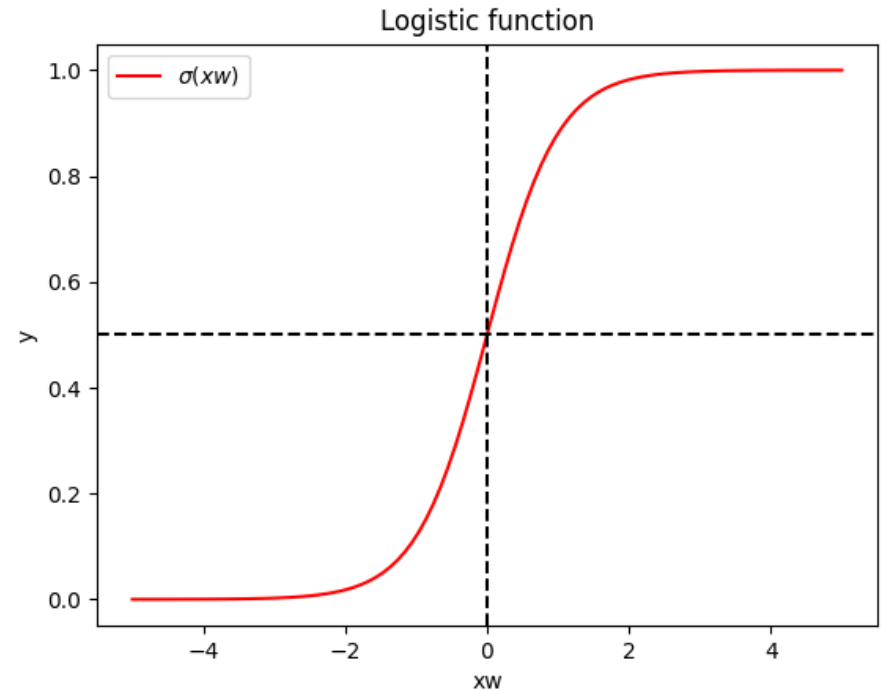
Default link function:

$$y \sim \text{Bernoulli}(p)$$

$$\mu = p = P(y = 1|x)$$

$$\ln\left(\frac{p}{1-p}\right) = X\beta$$

$$P(Y = y | x) = p^y(1-p)^{1-y}, \quad y \in \{0, 1\}$$



More than two categories

Poisson and negative binomial models are good options.

$$y \sim \text{Poisson}(\lambda)$$

$$\mu = \lambda = E(y \mid x)$$

$$\ln(\lambda) = X\beta$$

$$y \sim \text{NegativeBinomial}(\mu, r)$$

$$\mu = E(y \mid x)$$

$$\ln(\mu) = X\beta$$

$$P(Y = y \mid x) = \binom{y + r - 1}{y} \left(\frac{r}{r + \mu} \right)^r \left(\frac{\mu}{r + \mu} \right)^y, \quad y \in \{0, 1, 2, \dots\}$$

$$P(Y = y \mid x) = \frac{e^{-\lambda} \lambda^y}{y!}, \quad y \in \{0, 1, 2, \dots\}$$

where $r > 0$ is the **dispersion parameter** controlling overdispersion.

Statistical Testing

Learning objectives:

- Understand the formulation and assumption of standard statistical tests
- Understand the interpretation of a p-value
- Considerations when designing an experiment

Hypothesis Tests

Goal

- With a selected significant level, if two datasets show heterogeneity
- How to approach this by using **test statistics**

Many Types of Tests

- Parametric tests – **make assumptions on the data distribution**
 - Z-score vs. t-test
 - ANOVA (F-test)
- Non-parametric – **does not make assumptions on data dist.**
 - Permutation test

Hypothesis Formulation

Null Hypothesis

- Effect size is 0
- Difference between conditions is 0

Alternative Hypothesis

- Effect size is NOT 0
- Difference between conditions is NOT 0

We accept or reject the null based on the generated p-value (< 0.05 – this is self-selected)

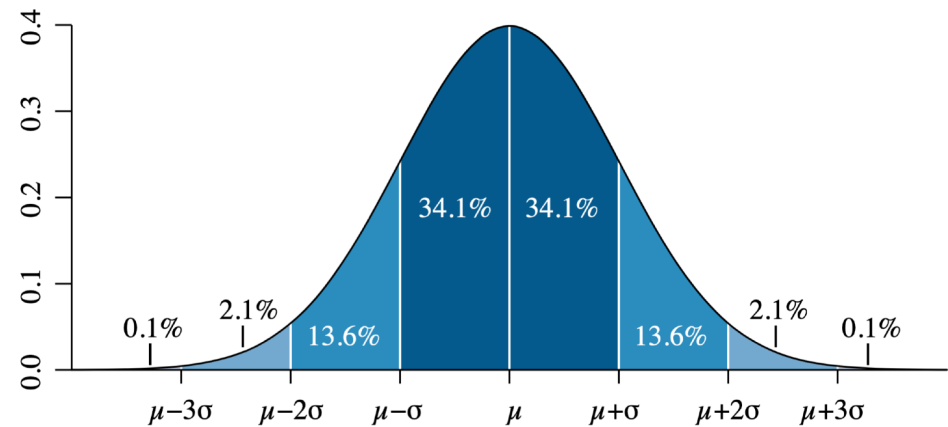
Understanding Significance

Statistical significance **DOES NOT** imply causality

It is a measurement of the likelihood of the observed data happening by chance.

A p-value tells us the probability of seeing the observation **if the null hypothesis is true.**

Z-score Test



Credit: Wikipedia

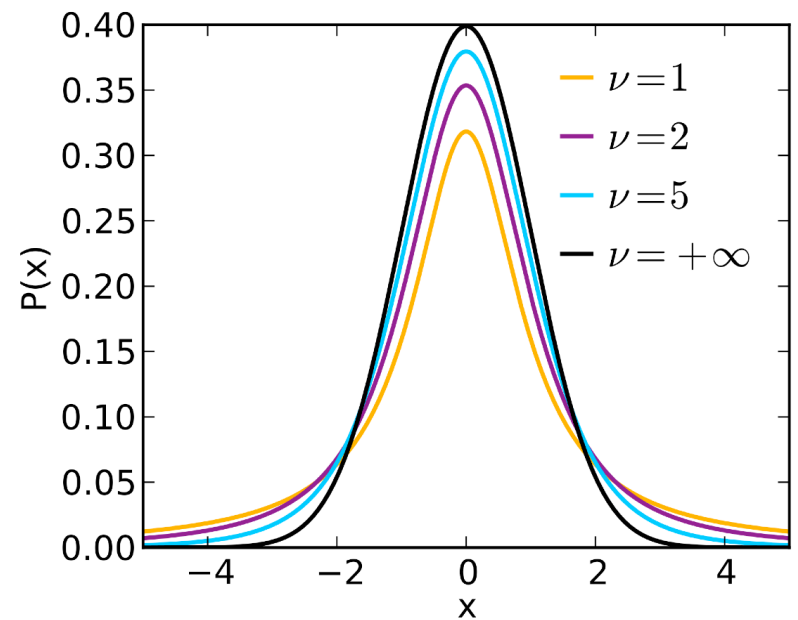
- Based on the Normal Distribution
- Generate a Z-score from our effect size estimate
- Used when you know your data follows a normal distribution or data size is large so that you can approximate a Normal distribution (central limit theorem)
- Compare to Normal(0,1) distribution
 - One-tail or Two-tail

$$Z = \frac{\beta - \hat{\beta}}{SE(\beta)}$$

t-test

Based on the student t distribution

- Defined by degrees of freedom (d.f)
 - $n-1$ d.f.
- More appropriate for smaller sample sizes (most cases)
- Use our sample mean $\bar{X}$ and sample s.d $\hat{\sigma}$ to calculate test statistic
- Value of μ depends on your null hypothesis
 - One-Sample - 0
 - Two-sample - μ_2



Credit: Wikipedia

$$t = \frac{\bar{X} - \mu}{s} = \frac{\bar{X} - \mu}{\hat{\sigma}/\sqrt{n}}$$

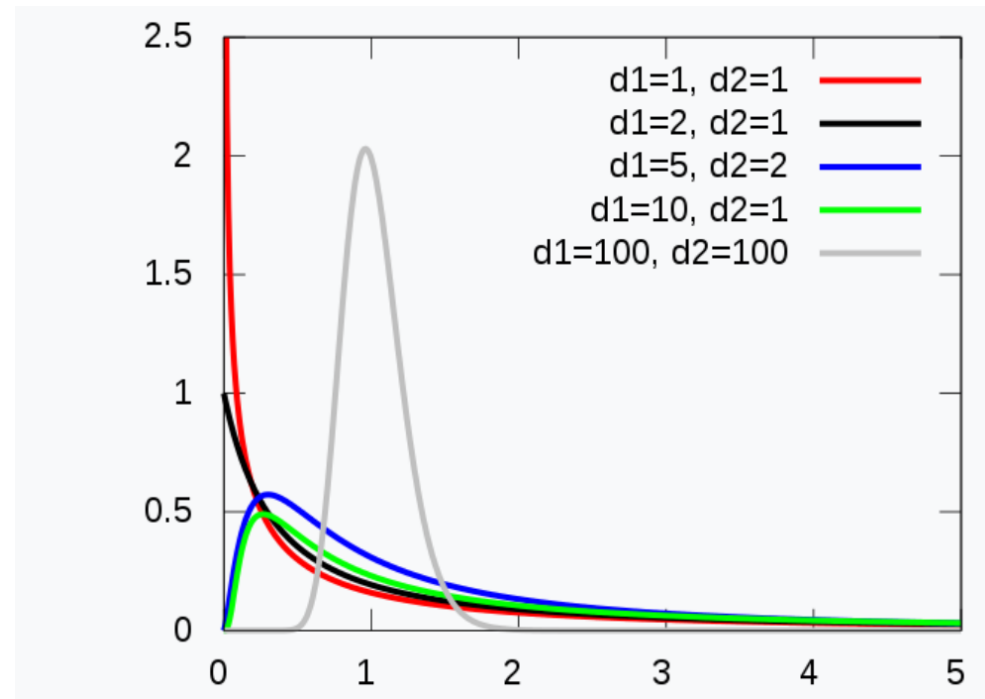
ANOVA

Based on the F-distribution

- Similar to t-statistic
- Statistic is the ratio of two Chi-squareds

Null hypothesis: multiple means are **all** equal

Alternative: At least one group is different



Credit: Wikipedia

ANOVA

How much variation is explained by our treatment condition?

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

Total sum of squares

$$SST = SS(Treatments) + SS(Residuals)$$

Take the ratio of the variance (mean sum of squares) between treatments over the variance within treatments

High variance between treatments will give a low p-value

Non-Parametric Test

Similar to what is done for enrichment analyses

- Avoids needing to make assumptions on distributions
 - Can be very **slow**
1. Shuffle labels in your data to generate estimates (random baseline)
 2. See where your true observation lies on this baseline

Multiple Testing

- May test multiple hypothesis (example, effect sizes in linear model)
- If null distribution is true, expect a **uniform distribution** of p-values
 - 5% of tests will give a significant result
- Multiple test correction methods are used to address this
 - Bonferonni Correction
 - **Benjamini & Hochberg Correction**
 - Found in results as **FDR, p-adjusted, q-value**

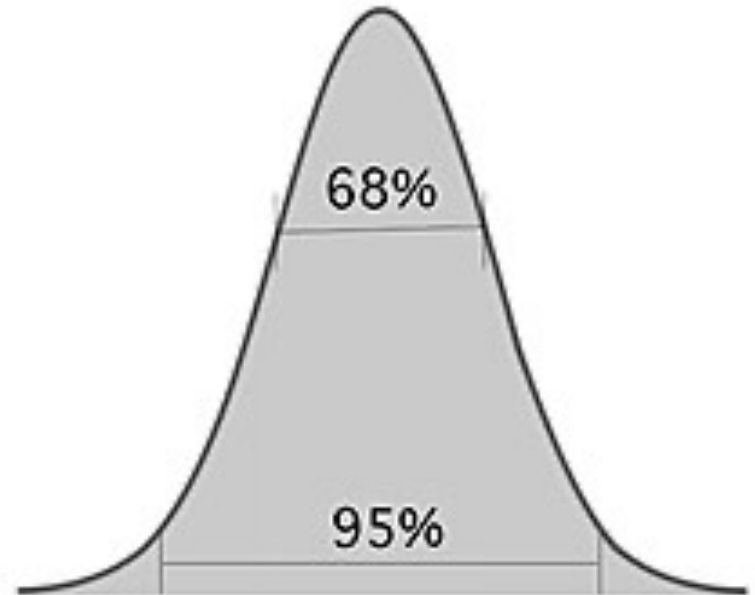
Confidence Intervals

- We may want to attribute a range to our estimate

- Can be calculated with the **margin of error**

$$E = Z \frac{\sigma}{\sqrt{n}}$$

- Where Z is the z-score that defines our confidence level

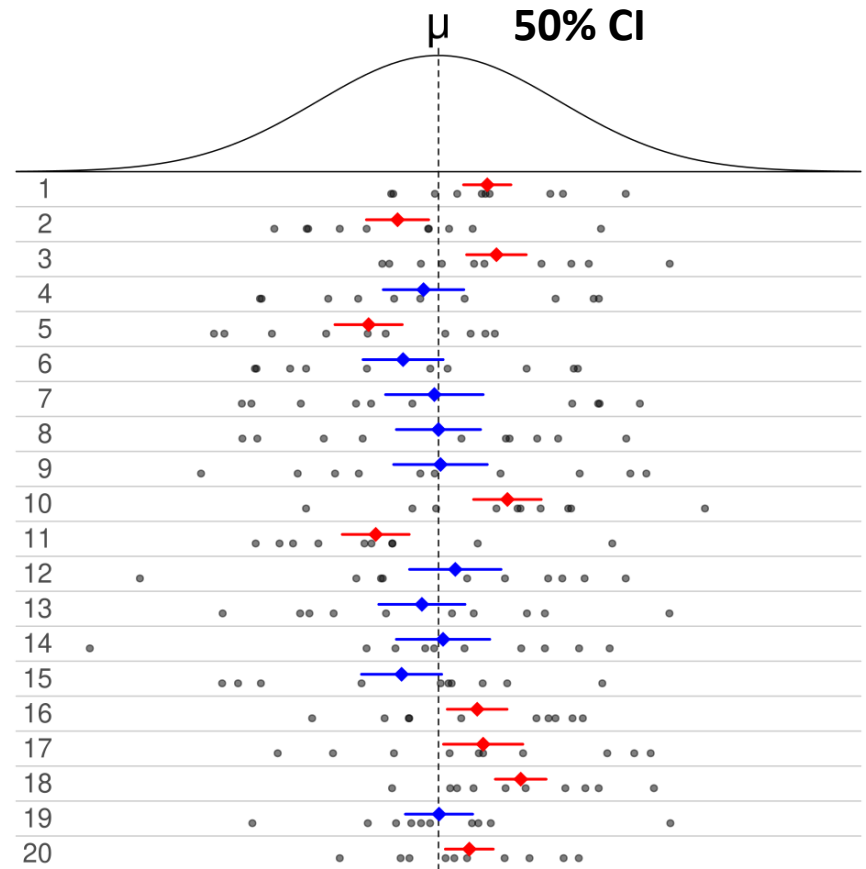


PSA: Confidence Intervals

A 95% CI **DOES NOT** mean there is a 95% chance the true value is in that range

For multiple experiments, 95% of CIs will overlap with the true value

Consider a Bayesian formulation instead (beyond scope)



Study Design

Learning objectives:

- Understand how simulations can be used to help with study design
- How to determine the required sample size

Study Design

Data collection can be expensive and time-consuming

Need to ensure that the collected data has enough **power** to avoid false negatives (type II error)

Can simulate data to help decide on the statistical test used and determine the appropriate **sample size**

Determining sample size

- Statistical test results depend on multiple factors
 - Difference in Samples
 - **Sample Size**
- We want to make sure we have enough samples to avoid False Negatives (type II errors)
- Can leverage the assumptions used for our statistical test to determine sample size

Confidence Intervals

In practice you will want your CI to cover a small range to be informative.
Can achieve this by solving for the **margin of error E**

$$E = Z \frac{\sigma}{\sqrt{n}}$$

$$n = \left(\frac{Z\sigma}{E} \right)^2$$

Note that we may not have a value for σ

- Use estimate from past study
- Make an assumptive guess
- Run a small pilot study

Sample Comparison

$$n_i = 2 \left(\frac{Z_{1-\frac{\alpha}{2}} + Z_{1-\beta}}{ES} \right)^2$$

$$\text{Effect Size (ES)} = \frac{|\mu_1 - \mu_2|}{\sigma}$$

$\alpha \rightarrow$ *Significance Level*

Probability of rejecting null when it is true

$1-\beta \rightarrow$ *Power*

Probability of rejecting null when it is false

To summarize

- ✓ Basics for data manipulation & cleaning
- ✓ Explored basic linear modeling for regression & classification
- ✓ Explored statistical testing and intuition behind study design

Now you are ready to:

- Use dplyr to facilitate reproducible data manipulation
- Apply and interpret linear model results
- Interpret confidence intervals and p-values
- Explore more complex models and tests!

Future Statistics Workshops

- Dimension Reduction
- **Bayesian Inference**
- Time-series Analysis
- And more...

Thank you for attending!



Scan the QR code to confirm you attended today's workshop.
Fill out the survey to help us improve our offerings!